

MOBILITY PRICING INTELLIGENCE: COST PREDICTION & PRICING DRIVER ANALYSIS

Pricing Drivers, Ride Economics &
Mobility Operations



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly &
ML libraries

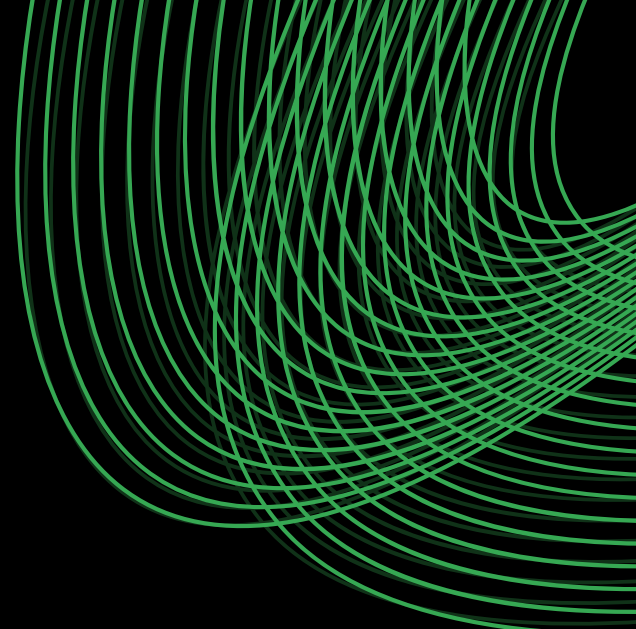
PRESENTED BY

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Overview & Business Objective



1.1 PROJECT OVERVIEW

Ride-sharing platforms operate in an environment where ride costs can vary based on several factors, including expected ride duration, vehicle type, rider demand, driver availability, booking period, location, and customer characteristics.

Understanding these relationships is important for estimating ride costs consistently and identifying the operational factors associated with pricing variation.

The Ride Pricing Analytics project applies machine learning techniques to historical ride-sharing data to develop a ride-cost prediction system and identify the factors that contribute most strongly to model predictions.

The project combines:

- Data understanding and validation
- Exploratory Data Analysis
- Feature engineering
- Data preprocessing
- Regression modeling
- Hyperparameter tuning
- Ensemble learning
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented interpretation

The objective is not only to predict historical ride costs, but also to understand why the models make their predictions and where significant prediction errors remain



1.2 PROBLEM STATEMENT



Ride costs can vary across different combinations of demand, supply, ride duration, vehicle type, booking period, location, and customer characteristics.

The company currently relies primarily on ride duration when determining ride fares. However, the historical dataset contains additional information that may provide useful signals for estimating ride costs.

The key problem addressed in this project is:

How can historical ride-sharing data be used to build a machine learning model that accurately predicts historical ride costs while identifying the major factors contributing to those predictions?

The project therefore treats Historical_Cost_of_Ride as the target variable and uses ride-level, customer, vehicle, booking, and demand-supply information as predictive features.

(Note: It is important to note that the dataset contains historical observed ride costs rather than an explicitly defined optimal or surge-pricing target)



1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Develop a machine learning-based ride-cost prediction system.
- Understand the underlying structure and characteristics of the data.
- Identify important pricing patterns through EDA.
- Analyze the relationship between ride duration & ride cost.
- Analyze demand and supply relationships using rider and driver information.
- Train and compare multiple regression models.
- Improve selected models using hyperparameter tuning.
- Evaluate Voting and Stacking ensemble approaches.
- Evaluate model performance using appropriate regression metrics.
- Investigate residuals and large prediction errors.
- Identify the most influential features driving model predictions.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, price modeling, model explainability, and business interpretation to understand demand patterns, evaluate predictive performance, identify key demand drivers, and derive actionable retail insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Distribution analysis
- Relationship analysis

02

Demand Modeling

- Feature preparation
- Regression modeling
- Model comparison
- Model evaluation

03

Explainability

- Lasso coefficient analysis
- SHAP analysis
- Permutation feature importance

04

Business Interpretation

- Business Interpretation
- Inventory implications
- Pricing and promotion observations
- Seasonal and weather-related effects
- Recommendations for retail planning



Data Understanding & Exploratory Data Analysis



2.1 DATA OVERVIEW

The dataset contains 1,000 observations and includes information related to riders, drivers, location, customer loyalty, previous rides, ratings, booking time, vehicle type, expected ride duration, and historical ride cost.

Data Source: Kaggle

Dataset Size

1,000

Records

10

Original Features

Purchase Diversity

3

Location Categories

2

Vehicle Types

4

Time Segments



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Number_of_Riders	Numerical (Discrete)	Number of riders associated with the observation
Number_of_Drivers	Numerical (Discrete)	Number of available drivers associated with the observation
Location_Category	Categorical	Geographic classification of the ride location
Customer_Loyalty_Status	Categorical	Loyalty category of the customer
Number_of_Past_Rides	Numerical (Discrete)	Number of previous rides completed by the customer
Average_Ratings	Numerical (Discrete)	Average rating associated with the customer
Time_of_Booking	Categorical	Time period during which the ride was booked
Vehicle_Type	Categorical	Category of vehicle selected for the ride
Expected_Ride_Duration	Numerical (Discrete)	Expected duration of the ride
Historical_Cost_of_Ride	Numerical (Continuous)	Historical cost of the ride and target variable used for regression modeling

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Historical_Cost_of_Ride

Historical_Cost_of_Ride represents the observed historical cost associated with each ride.

The target has:

372.50

Mean

362.02

Median

25.99

Minimum

836.12

Maximum

187.16

Standard Deviation

The target demonstrates substantial variation across observations, making it suitable for a continuous regression problem.



2.3 DATA QUALITY & PREPARATION OVERVIEW

The initial data-quality analysis identified the following characteristics:

Missing Values

No missing values were identified across the dataset.

Duplicate Records

No duplicate records were identified.

Numerical Values

All numerical variables contained valid non-negative observations.

Categorical Variables

The categorical variables contained well-defined categories without unexpected values or extremely rare categories requiring consolidation.

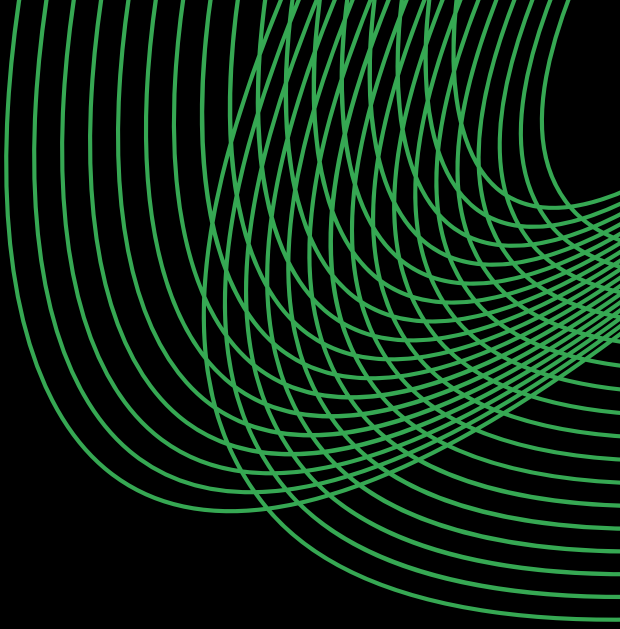
Outlier Analysis

An IQR-based analysis identified 10 potential outliers in Number_of_Drivers, representing 1.0% of the dataset.

Investigation showed that these observations contained driver counts between 80 & 89 and were accompanied by valid values across the remaining variables.

Therefore, these observations were considered legitimate high-supply observations rather than data errors and were retained for modeling.





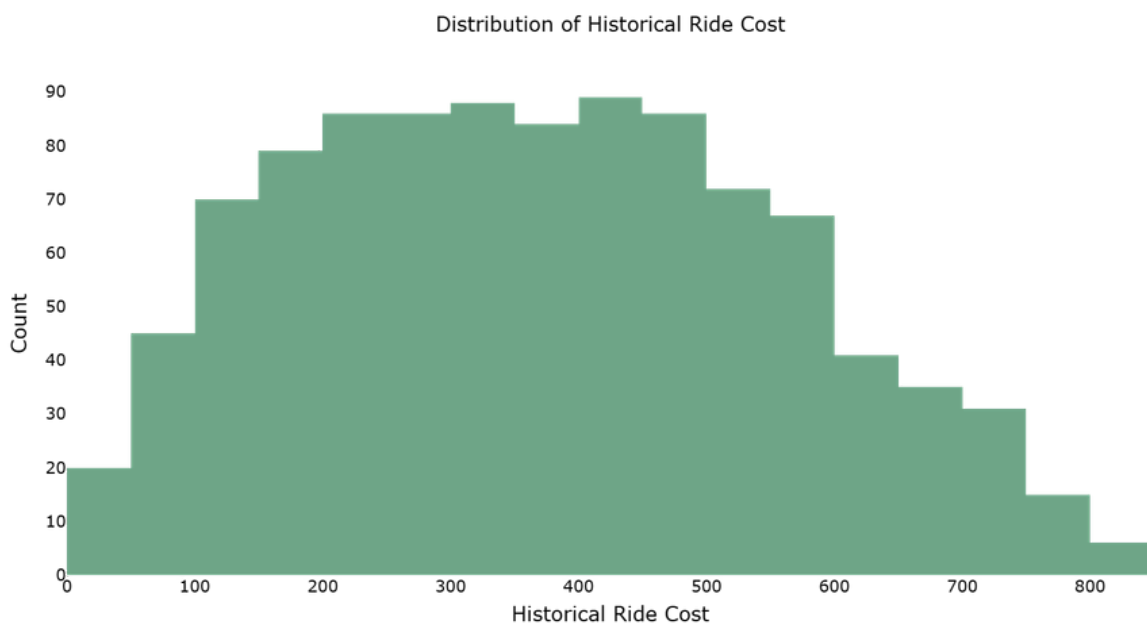
2.4 Exploratory Data Analysis



Historical ride costs are broadly distributed across a wide range, with the highest concentration occurring in the mid-range.

\$2,135

Average Cost per Service



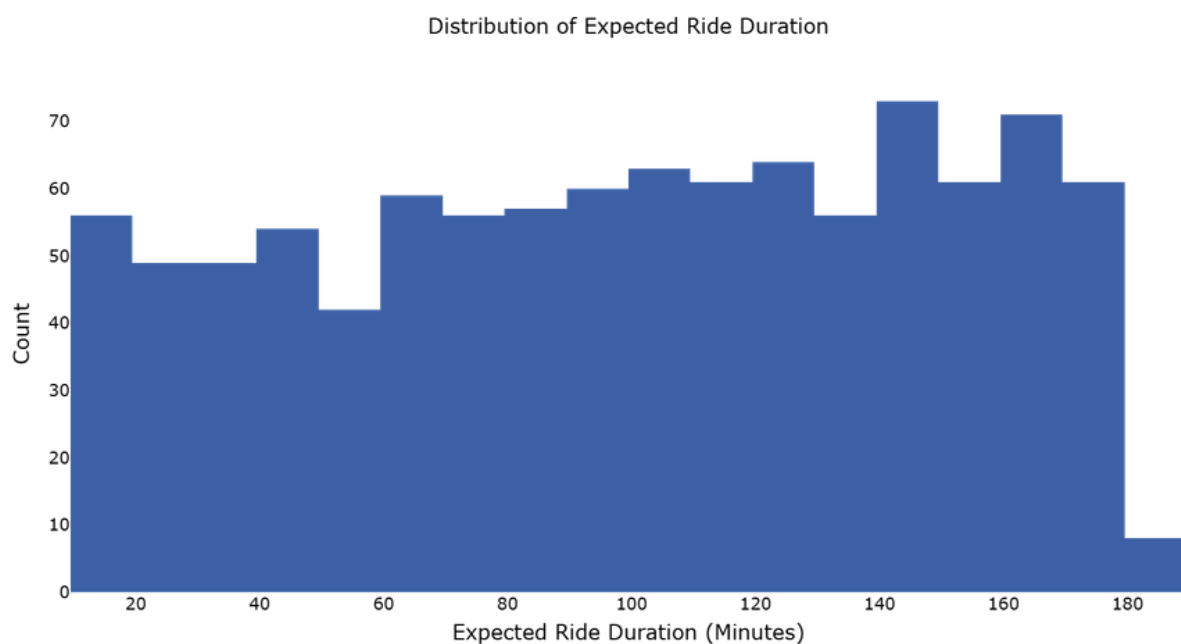
Key observations

- Historical ride costs span approximately \$25 to \$836, with the highest concentration of observations occurring roughly between \$150 and \$500.
- The distribution is reasonably broad, with observations extending across both lower-cost and higher-cost rides rather than being concentrated around a single narrow price range.

Business Insights

- The wide cost range indicates that the pricing model needs to account for multiple ride characteristics rather than relying on a single fixed fare level.
- The concentration of rides in the mid-cost range provides a substantial base of observations for learning typical pricing patterns, while higher-cost rides require the model to handle greater variation.

Expected ride durations are distributed across a broad operating range, with observations spread relatively consistently across most durations.



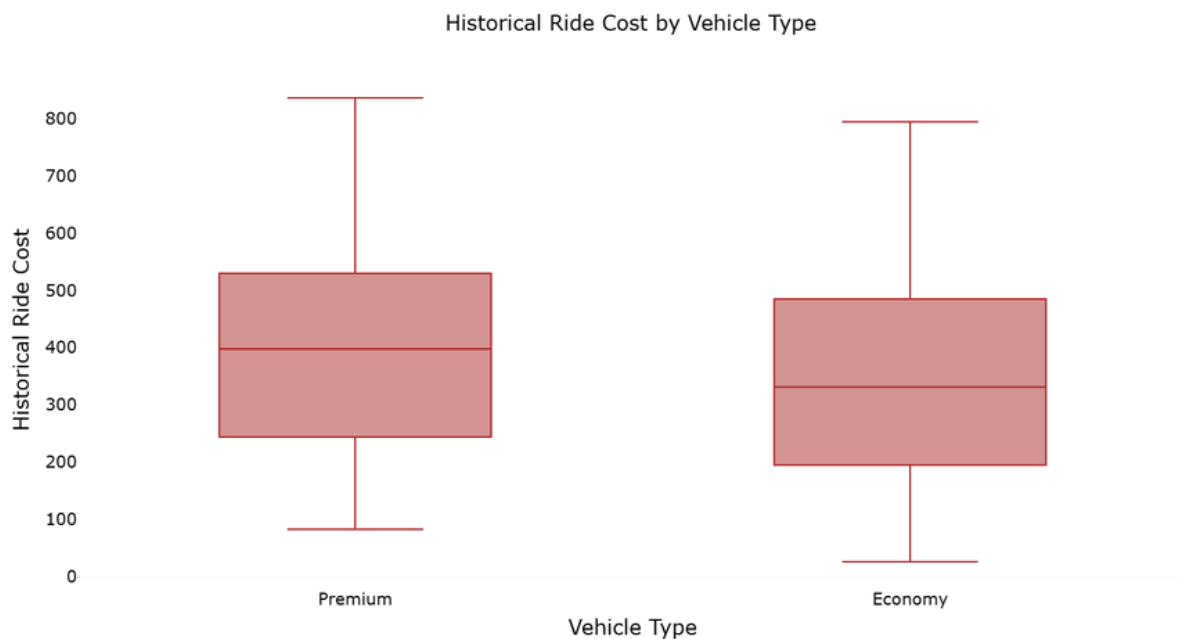
Key observations

- Expected ride duration ranges from approximately 10 to 180 minutes, with observations distributed across most of this range.
- The distribution does not show a strong concentration around one duration; several intervals between approximately 60 and 175 minutes contain relatively high observation counts.

Business Insights

- The broad duration range provides the model with observations covering both shorter and longer rides, supporting the analysis of duration-related pricing patterns.
- Since ride duration varies substantially across trips, incorporating Expected_Ride_Duration into the pricing model is important for capturing differences in historical ride costs.

Premium rides generally have higher historical costs than Economy rides, while both vehicle types show substantial cost variation.



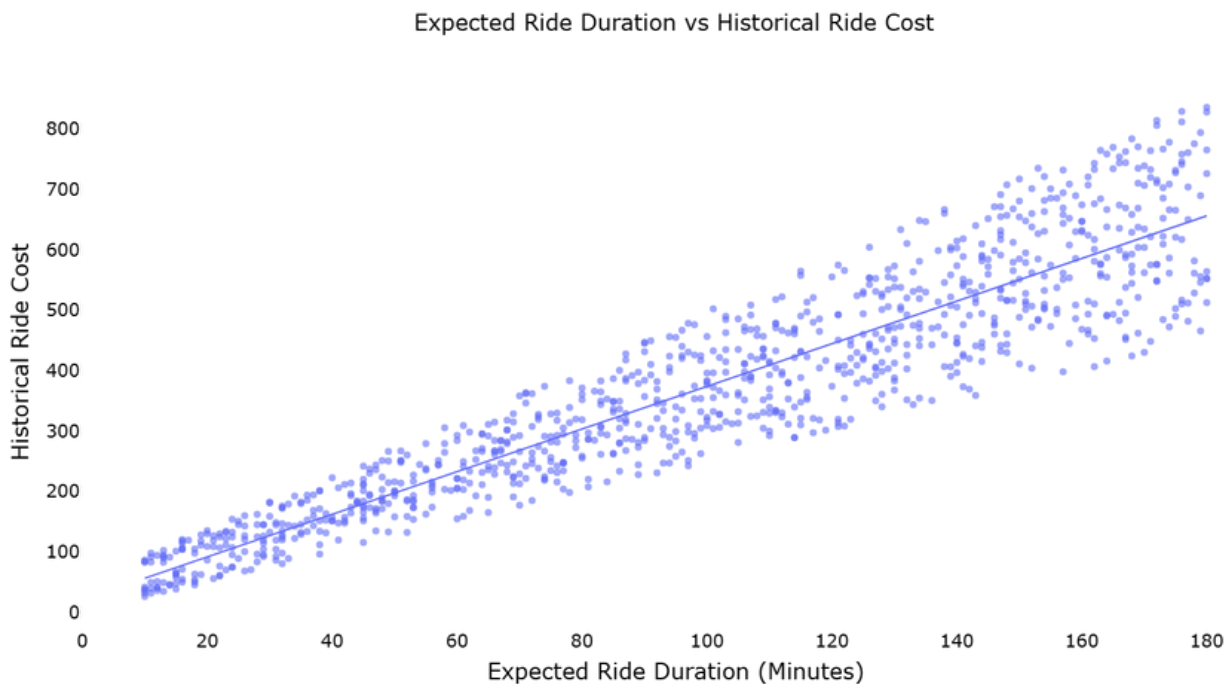
Key observations

- The median historical cost for Premium rides is visibly higher than the median for Economy rides, with the Premium median around \$400 compared with roughly \$330 for Economy.
- Both vehicle types have wide cost ranges, extending from relatively low-cost rides to values approaching or exceeding \$800, indicating that vehicle type alone does not explain the full variation in historical ride costs.

Business Insights

- Vehicle type should be incorporated into pricing models because the observed cost distributions differ between Premium and Economy rides.
- The substantial variation within each vehicle category indicates that pricing decisions should combine vehicle type with other ride characteristics rather than applying a fixed vehicle-specific price.

Historical ride cost shows a strong positive relationship with expected ride duration



Key observations

- Historical ride cost increases consistently as expected ride duration increases, with the plotted relationship showing a strong positive trend across the full duration range.
- The spread of ride costs becomes wider at longer durations, particularly beyond approximately 120 minutes, indicating greater variation in historical costs for longer rides.

Business Insights

- Expected ride duration should be an important input in ride-cost prediction because longer rides are consistently associated with higher historical costs.
- The increasing variation at longer durations indicates that additional ride characteristics are needed to explain differences in cost beyond duration alone.

Driver availability generally increases as the number of riders increases, while supply variation becomes wider at higher rider volumes



Key observations

- The scatter plot shows a clear positive relationship between the number of riders and the number of drivers, indicating that higher rider volumes are generally accompanied by greater driver availability.
- At higher rider counts, the range of observed driver counts becomes substantially wider, showing greater variation in available supply under higher demand conditions.

Business Insights

- Pricing analysis should consider both rider demand and driver supply rather than treating either variable independently.
- The variation in driver availability at higher rider volumes supports the use of a combined demand-supply feature such as Rider_Driver_Ratio to represent the balance between riders and drivers.

Historical ride costs show relatively similar distributions across booking periods, with only moderate differences in median cost.

52.03%

Typical CPU Utilization (Median)



Key observations

- The median historical cost is highest for Afternoon bookings, followed closely by Morning, while Night has the lowest median among the four periods.
- The distributions overlap substantially across all four booking periods, and each period shows a wide range of historical ride costs.

Business Insights

- Time of booking can be retained as a pricing-related feature, but the substantial overlap suggests that booking period alone does not determine historical ride cost.
- Pricing analysis should combine booking time with stronger ride-level predictors such as expected duration and vehicle type rather than applying substantially different pricing solely by time period.

The positive duration-cost relationship remains consistent across vehicle types and locations, with Premium rides generally positioned above Economy rides

51.63%

Typical Memory Utilization (Median)



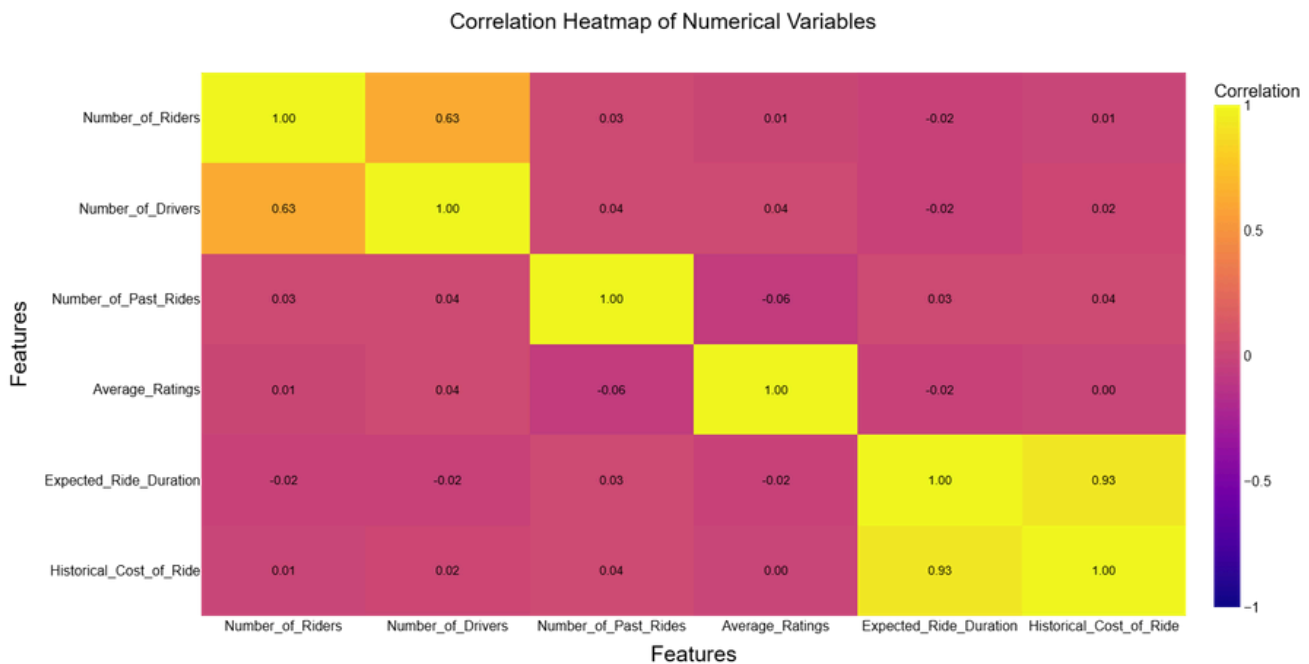
Key observations

- Historical ride cost increases with expected ride duration across Urban, Suburban, and Rural locations, showing a consistent positive relationship in all three panels.
- Within each location, Premium rides generally show higher historical costs than Economy rides at comparable ride durations, although the two groups overlap.

Business Insights

- Ride duration, vehicle type, and location should be considered together when analyzing historical ride costs rather than evaluating each factor independently.
- The consistent pattern across locations suggests that the relationship between duration and historical cost is not limited to a single geographic category.

Expected ride duration has the strongest numerical relationship with historical ride cost, while rider and driver counts show a moderate relationship with each other.

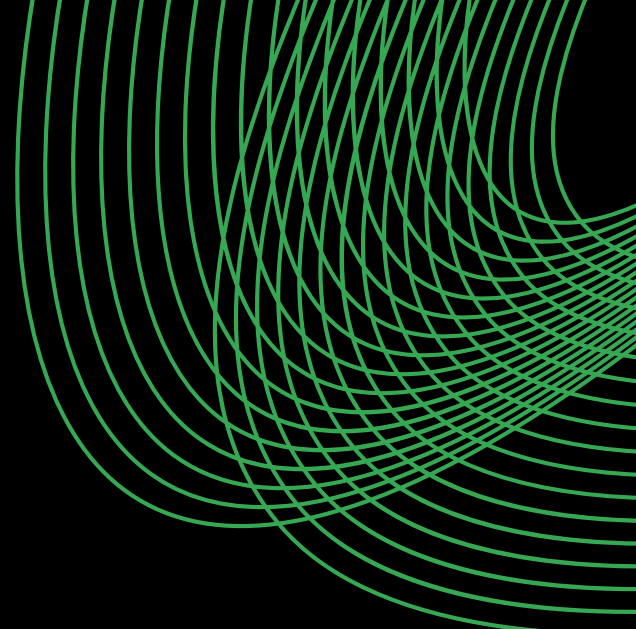


Key Observations

- Expected_Ride_Duration has a very strong positive correlation with Historical_Cost_of_Ride of approximately 0.93, making it the strongest numerical relationship with the target.
- Number_of_Riders and Number_of_Drivers have a moderate positive correlation of approximately 0.63, while the remaining numerical variables show correlations close to zero with historical ride cost.

Business Insights

- Expected ride duration should receive substantial attention in ride-cost modeling because of its strong observed relationship with historical ride cost.
- The moderate rider-driver relationship supports examining demand and supply together, which is consistent with the engineered Rider_Driver_Ratio feature used during model preparation.



Data Preparation & Feature Engineering



3.1 DATA PREPARATION

Following the exploratory analysis, the dataset was prepared for machine learning.

The preparation process included:

- Selecting the target variable.
- Separating predictor variables from the target.
- Identifying numerical and categorical variables.
- Creating the demand-supply feature.
- Splitting the dataset into training and testing portions.
- Preparing numerical variables for scaling.
- Preparing categorical variables for one-hot encoding.
- Saving the processed datasets and preprocessing artifact for subsequent notebooks.

The target variable was:

Historical_Cost_of_Ride

The final dataset was divided using an 80/20 train-test split.

- Training observations: 800
- Testing observations: 200

A random_state of 42 was used to ensure reproducibility.



3.2 DEMAND-SUPPLY FEATURE ENGINEERING

The EDA demonstrated a relationship between the number of riders and available drivers.

To represent demand relative to available supply, the following feature was created:

Rider_Driver_Ratio

The feature was calculated as:

Rider_Driver_Ratio = Number_of_Riders / Number_of_Drivers

This feature provides a direct representation of the number of riders relative to available drivers.

The feature was retained as an additional numerical predictor for modeling



3.3 CATEGORICAL ENCODING

The dataset contains four categorical variables:

- *Location_Category*
- *Customer_Loyalty_Status*
- *Time_of_Booking*
- *Vehicle_Type*

These variables were transformed using One-Hot Encoding.

The encoder was configured with `handle_unknown='ignore'` to ensure that previously unseen categories would not cause errors during prediction.

3.4 NUMERICAL FEATURE SCALING

The numerical variables were standardized using `StandardScaler`.

- *Number_of_Riders*
- *Number_of_Drivers*
- *Number_of_Past_Rides*
- *Average_Ratings*
- *Expected_Ride_Duration*
- *Rider_Driver_Ratio*

The scaler was fitted using the training data and then applied to both training and testing data.



3.5 TRAIN-TEST SPLIT

The prepared dataset was divided into:

- 80% training data: 800 observations
- 20% testing data: 200 observations

The training data was used for model fitting and hyperparameter optimization, while the testing data was reserved for final evaluation.

After preprocessing and one-hot encoding, the dataset contained 18 processed features.

The processed datasets were saved as:

- X_train.csv
- X_test.csv
- y_train.csv
- y_test.csv

The fitted preprocessing pipeline was also saved for reuse during model evaluation and explainability.





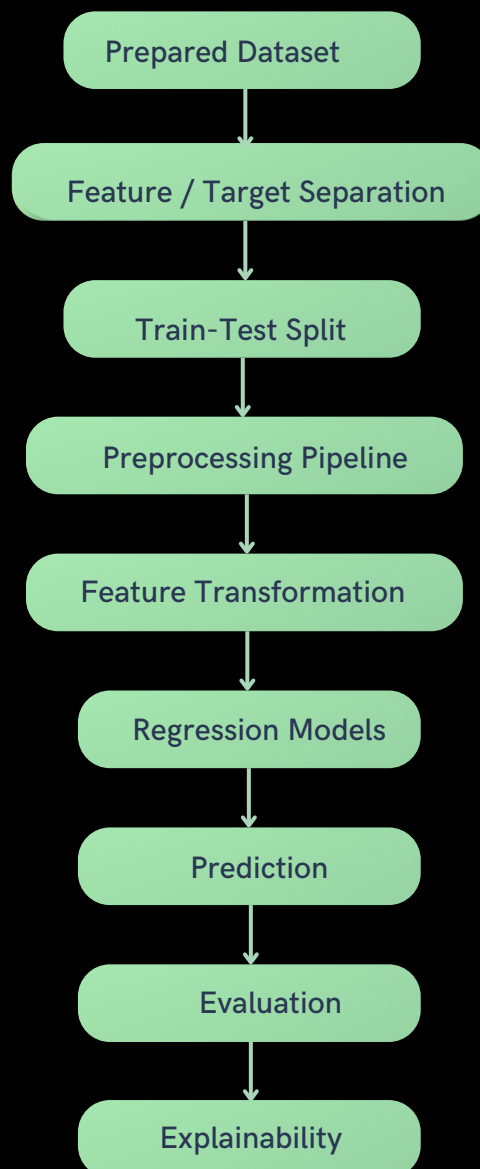
Model Development & Experimentation



4.1 MODELING APPROACH

The problem was formulated as a supervised regression problem, with *Historical_Cost_of_Ride* as the continuous target variable.

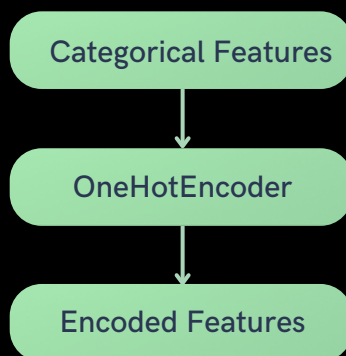
The modeling workflow consisted of:



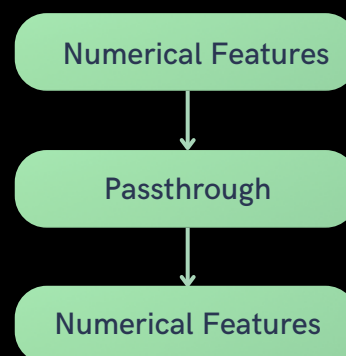
4.2 PREPROCESSING PIPELINE

A unified preprocessing pipeline was used to ensure that the same transformations were applied consistently during training and prediction.

Categorical Pipeline



Numerical Pipeline



Both were combined through a *ColumnTransformer*.

The fitted preprocessor was saved as a reusable artifact to maintain consistency across subsequent evaluation and explainability notebooks.



4.3 REGRESSION MODELING

Multiple regression approaches were developed and evaluated.

Initial Models

- Linear Regression
- Ridge Regression
- Lasso Regression
- Random Forest
- Extra Trees
- Gradient Boosting
- MLP Regressor
- XGBoost
- LightGBM

Tuned Models

Hyperparameter optimization was performed for:

- Random Forest
- Extra Trees
- Gradient Boosting
- XGBoost
- LightGBM



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A lightweight RandomizedSearchCV configuration was used with:

- 8 parameter combinations
- 3-fold cross-validation
- 24 fits per model
- n_jobs=1
- random_state=42

This configuration was selected to provide meaningful hyperparameter exploration while keeping computational requirements practical.

Ensemble Models

Two ensemble approaches were also developed:

- Voting Regression
- Stacking Regression

Two ensemble approaches were also developed:

- Ridge Regression
- Gradient Boosting
- XGBoost
- LightGBM

with a final Ridge Regression estimator.

Baseline Model

A mean-based DummyRegressor was used as the baseline.

- MAE: 160.9642
- RMSE: 191.1528
- R^2 : -0.0022

This baseline provided a reference point for determining whether the developed models learned useful predictive relationships.



Model Evaluation & Comparison



5.1 EVALUATION METRICS

The developed models were evaluated using three primary regression metrics.

Mean Absolute Error (MAE)

- MAE measures the average absolute difference between actual and predicted historical ride costs.
- A lower MAE indicates that predictions are closer to actual values on average.

Root Mean Squared Error (RSME)

- RMSE gives greater weight to larger prediction errors.
- This makes RMSE useful for identifying models that produce relatively large prediction errors.

R² Score

- R² measures the proportion of variation in historical ride costs explained by the model.
- A higher R² indicates stronger explanatory performance on the evaluated dataset.



5.2 ERROR ANALYSIS

Beyond the overall evaluation metrics, prediction errors were examined through:

- Residual analysis
- Actual vs predicted analysis
- Baseline comparison
- Maximum absolute error analysis
- Model-level error comparison

The evaluation was performed on the held-out test dataset containing 200 observations.



5.2.1 RESIDUAL ANALYSIS

Residuals were calculated as:

$$\text{Actual} - \text{Predicted}$$

This indicates that model performance varies depending on product category.

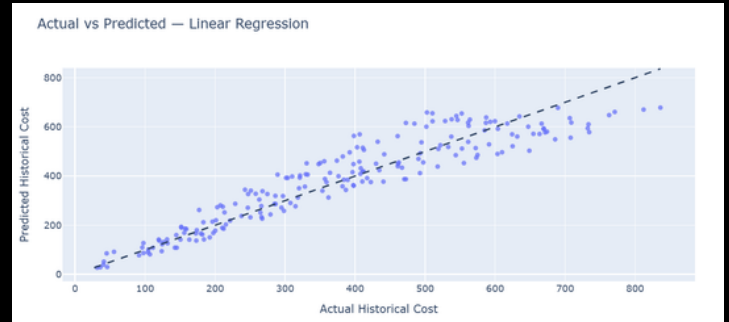
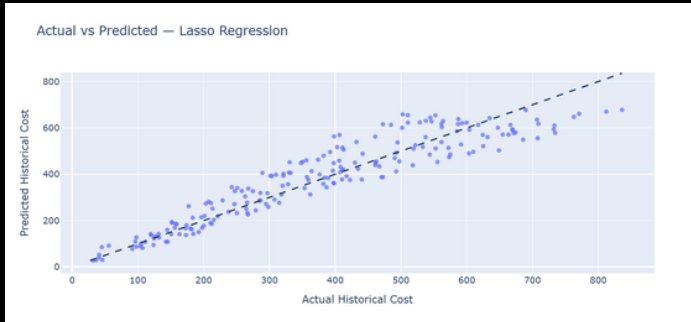
The negative mean errors observed across the models indicate a slight tendency toward over-prediction.

However, individual residuals were considerably larger for some observations.

The maximum absolute errors varied across models, demonstrating that strong aggregate performance does not eliminate large errors for individual rides.



5.2.2 ACTUAL VS PREDICTED ANALYSIS



The actual-versus-predicted analysis showed that predictions generally align with observed historical ride costs.

Most observations were positioned relatively close to the reference line, indicating that the models captured a substantial proportion of the variation in ride costs.

However, deviations from the reference line remained visible for individual observations, demonstrating that some ride costs were more difficult to estimate accurately.



5.3 LARGEST PREDICTION ERRORS

The largest individual error investigated in detail occurred for the tuned Gradient Boosting model.

Metric	Value
Actual Historical Cost	398.32
Predicted Historical Cost	598.44
Absolute Error	200.12

The model therefore over-predicted the historical ride cost by approximately 200.12 for this observation.

The corresponding ride characteristics were:

- Number of Riders: 59
- Number of Drivers: 40
- Location Category: Rural
- Customer Loyalty Status: Gold
- Number of Past Rides: 64
- Average Ratings: 4.72
- Time of Booking: Night
- Vehicle Type: Economy
- Expected Ride Duration: 157 minutes
- Rider_Driver_Ratio: 1.475
- Actual Cost: 398.32
- Predicted Cost: 598.44



5.4 INVESTIGATION OF THE LARGEST ERROR

The selected observation represents a Rural, Economy-vehicle ride booked at Night.

- 59 riders
- 40 drivers
- A rider-driver ratio of 1.475
- 64 past rides
- An average rating of 4.72
- An expected ride duration of 157 minutes

Despite the long expected ride duration, the historical ride cost was only approximately 398.32.

The tuned Gradient Boosting model predicted approximately 598.44.

Explainability Conclusion

The model's prediction was strongly influenced by expected ride duration, resulting in a substantial over-prediction for this particular ride.

- The individual SHAP explanation showed that Expected_Ride_Duration contributed approximately +226.52 to the model prediction.
- Other features provided smaller positive and negative adjustments, but their combined contributions did not sufficiently offset the strong contribution associated with expected ride duration.
- This demonstrates that the model learned a strong relationship between expected ride duration and historical ride cost, but that relationship does not perfectly describe every individual observation.



5.5 UNTUNED VS TUNED MODEL COMPARISON

Hyperparameter tuning produced mixed results across the five non-linear model families.

Model	Untuned RMSE	Value
Random Forest	74.6436	78.9125
Extra Trees	72.0363	83.8946
Gradient Boosting	72.1217	69.9215
XGBoost	71.9309	73.5117
LightGBM	79.1346	72.1934

Interpretation

Gradient Boosting and LightGBM achieved lower test-set RMSE after tuning.

Random Forest, Extra Trees, and XGBoost produced higher test-set RMSE after tuning.

This demonstrates that hyperparameter tuning does not automatically improve performance on unseen data. The effectiveness of tuning depends on the model, search space, and selected parameter configuration.



5.6 ENSEMBLE MODEL COMPARISON

The ensemble models demonstrated different performance characteristics.

Voting Regression

- MAE: 52.9107
- RMSE: 69.6303
- R^2 : 0.8670

Voting Regression

- MAE: 52.0649
- RMSE: 67.7824
- R^2 : 0.8740

Stacking Regression achieved a lower MAE than the individual models included in the ensemble, while its RMSE remained close to the strongest individual models.

Voting Regression produced a higher RMSE than the strongest individual models.

The results demonstrate that combining multiple models does not automatically produce lower error across every evaluation metric.



5.7 OVERALL MODEL COMPARISON

The final evaluation included 16 trained models.

Model	MAE	RMSE	R ²
Lasso Regression	52.6056	67.5878	0.8747
Linear Regression	52.6138	67.5885	0.8747
Ridge Regression	52.5943	67.5893	0.8747
Stacking Regression	52.0649	67.7824	0.8740
Voting Regression	52.9107	69.6303	0.8670
MLP Regressor	55.5282	69.8909	0.8660
Gradient Boosting Tuned	53.3413	69.9215	0.8659
XGBoost	54.2290	71.9309	0.8581
Extra Trees	53.0204	72.0363	0.8577
Gradient Boosting	53.6769	72.1217	0.8573
LightGBM Tuned	54.7664	72.1934	0.8571



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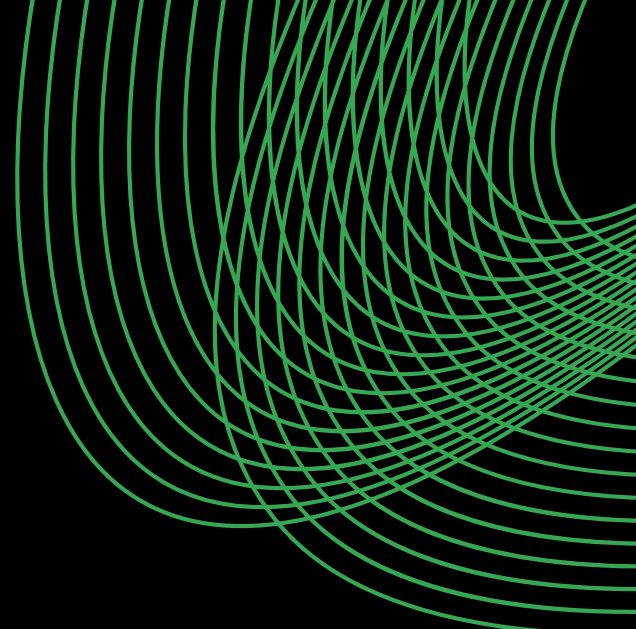
Model	MAE	RMSE	R ²
XGBoost Tuned	54.7342	73.5117	0.8518
Random Forest	55.6138	74.6436	0.8472
Random Forest Tuned	63.3565	78.9125	0.8292
LightGBM	57.9706	79.1346	0.8282
Extra Trees Tuned	66.9269	83.8946	0.8070

The strongest-performing models were closely grouped across the three evaluation metrics.

- **Lowest MAE:** Stacking Regression - 52.0649
- **Lowest RMSE:** Lasso Regression - 67.5878
- **Highest R²:** Lasso Regression - 0.8747

Therefore, no single model records the lowest error across all three evaluation metrics.





Explainability & Final Model



6.1 WHY EXPLAINABILITY WAS INCLUDED

Accurate predictions alone are not always sufficient for business applications.

Retail decision-makers also need to understand:

- Which variables influence predictions?
- Which ride characteristics matter most?
- What drives model behavior?
- Where does the model struggle?
- Why can individual predictions differ substantially from actual historical costs?

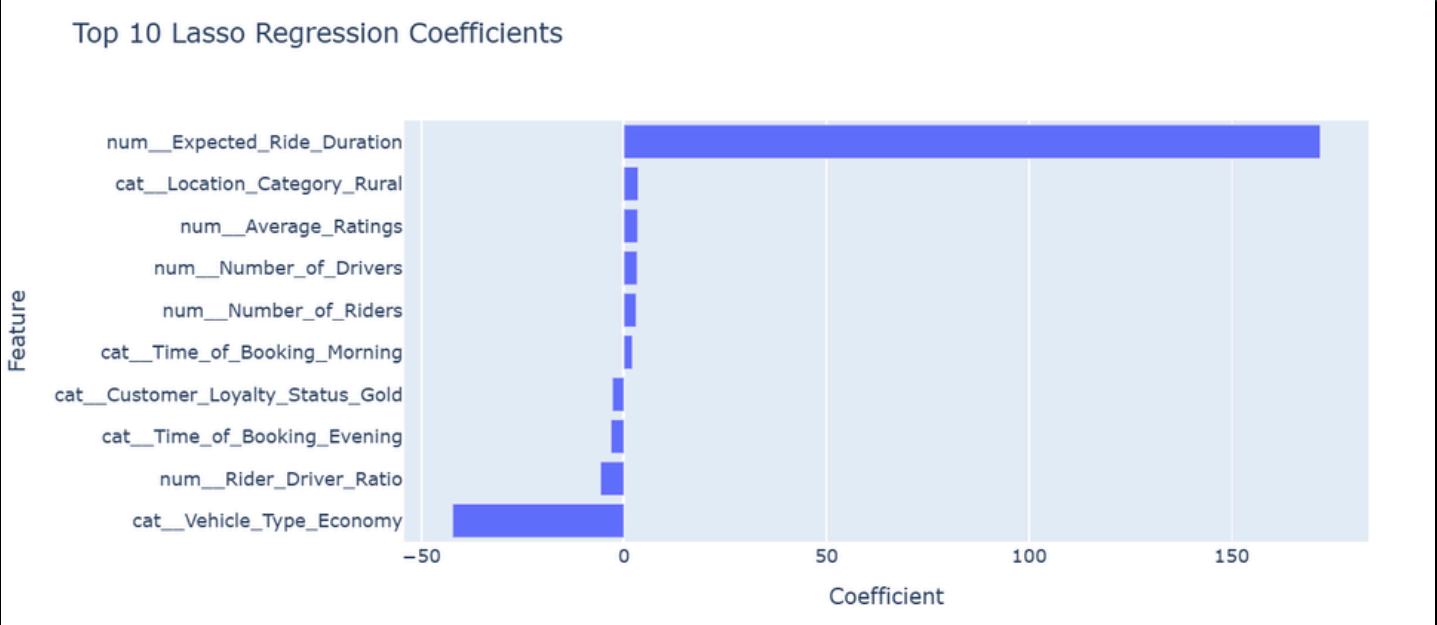
Explainability analysis was therefore performed to understand the model's prediction behavior.

The analysis included:

- Lasso Regression coefficients
- SHAP analysis
- SHAP directional analysis
- Permutation feature importance
- Individual prediction analysis
- Stacking component analysis



6.2 FEATURE IMPORTANCE



The highest-ranked model features were:

Feature	Value
Expected_Ride_Duration	34.22
Vehicle_Type_Economy	22.52
Rider_Driver_Ratio	11.25
Location_Category_Rural	3.47
Average_Ratings	2.41
Number_of_Drivers	2.40



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Feature	Value
Time_of_Booking_Evening	34.22
Number_of_Riders	22.52
Customer_Loyalty_Status_Gold	11.25
Time_of_Booking_Morning	3.47

Expected_Ride_Duration has the largest absolute coefficient by a substantial margin.

Lasso also assigns zero or effectively zero coefficients to several encoded features, including:

- Location_Category_Suburban
- Customer_Loyalty_Status_Regular
- Vehicle_Type_Premium

This demonstrates the feature-selection effect of Lasso regularization.

The coefficient values describe the model's learned relationships within the processed feature space and should not be interpreted as causal effects.



6.3 SHAP FEATURE IMPORTANCE

SHAP analysis was performed on the tuned Gradient Boosting model.

The highest mean absolute SHAP values were:

Rank	Feature	Value
1	Expected_Ride_Duration	150.3693
2	Vehicle_Type_Premium	10.4511
3	Vehicle_Type_Economy	5.9326
4	Rider_Driver_Ratio	3.8213
5	Number_of_Drivers	3.6967
6	Average_Ratings	3.5726
7	Number_of_Riders	1.1874
8	Number_of_Past_Rides	0.8486
9	Location_Category_Rural	0.4876
10	Customer_Loyalty_Status_Gold	0.3380

Expected_Ride_Duration is clearly the dominant feature, with a mean absolute SHAP value of approximately 150.37.

Vehicle type provides the next notable contribution, followed by demand-supply and other operational characteristics.

